Mahmoud Ahmed Abdelghani

Computer Engineering & Systems Student | Flutter / Mobile Developer

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Professional Summary

Senior Computer and Systems Engineering student at Zagazig University and a dedicated Flutter Developer with a problem-solver mindset. My strong foundation in algorithms, data structures, and core software engineering principles allows me to learn and apply Flutter and Dart deeply and intelligently. Navigating the rigorous demands of engineering school has naturally equipped me with essential soft skills, including performing effectively under pressure, setting strict deadlines for myself, and prioritizing tasks efficiently. While my primary passion and expertise are focused on mobile software development, my academic journey has also provided me with a solid, complementary understanding of hardware and embedded systems.

Education

Zagazig University, Faculty of Engineering **Egypt**

Bachelor of Science in Computer Engineering and Systems (Cumulative GPA: Very Good) *Expected Graduation: 2027*

- **Relevant Coursework:** Data Structures, Algorithms, Database Systems, Embedded Systems, Microprocessors, Control Systems, Digital Logic Design

Technical Skills

Programming Languages: Dart, Java, C++, C, Python, JavaScript, HTML, CSS
Mobile Development: Flutter SDK, Dart, Responsive UI/UX, Animations, Material/Cupertino Design
State Management & Architecture: BLoC, Cubit, GetX, Riverpod, Clean Architecture, SOLID Principles, OOP, Design Patterns
Backend & APIs: RESTful APIs, Dio, Interceptors, Firebase (Auth, Firestore, FCM, Storage), Supabase
Problem Solving & AI: Competitive Programming, Data Structures, Algorithms, Generative AI, LLMs, Prompt Engineering
Databases & Local Storage: SQL, MySQL, NoSQL, Secure Storage, Shared Preferences, Hive
Tools & Version Control: Git, GitHub, Android Studio, VS Code, Figma, Postman, Dependency Injection (GetIt), Linux

Featured Projects

Ma3refa Mobile | *Flutter, BLoC/Cubit, Dio, Secure Storage, Figma* **GitHub Link**

- Built an interactive AI-powered quiz and personalized learning application from scratch in 10 days.
- Implemented quiz flows and state management using BLoC/Cubit.
- Integrated Dio for API communication, GetIt for dependency injection, and Secure Storage for local persistence.
- Designed the complete UI/UX in Figma with a focus on usability and accessibility.

Sampark Chat App | *Flutter, GetX, Firebase (Auth, Firestore, Cloud Messaging)* **GitHub Link**

- Built a real-time messaging platform supporting one-to-one and group chats with multimedia sharing.
- Implemented authentication and group-chat features using Firebase Authentication and Firestore.
- Integrated Cloud Messaging and Storage and used GetX for state management.

Laza - E-Commerce App | *Flutter, BLoC/Cubit, Firebase* **GitHub Link**

- Developed a full-featured e-commerce mobile application with product browsing, cart, wishlist, reviews, and ratings.
- Integrated Firebase Authentication and Cloud Firestore and implemented local caching for improved performance.
- Built responsive mobile interfaces with Flutter and BLoC/Cubit.

Laser Setter | *Flutter, C++, Arduino, Bluetooth* **GitHub Link**

- Created a Flutter control interface for a hardware-based laser leveling system.
- Sent real-time angular data to servo motors through Bluetooth serial communication.
- Developed Arduino firmware in C++ for hardware control and sensor integration.

File System Simulation | *Java, JavaFX, Trees, Linked Lists* **GitHub Link**

- Simulated file-system structures and operations using Trees and Linked Lists.
- Implemented file creation, deletion, navigation, and metadata management with Java, JavaFX, and Gson.

Training & Experience

Code Plus

Advanced Flutter Trainee

- Completed a 40-hour advanced training covering Clean Architecture, REST APIs deep dive with interceptors, Dependency Injection (DI), and state management techniques (BLoC, Riverpod).
- Integrated Backend-as-a-Service solutions including Firebase and Supabase.
- Contributed to "Doctor Hunt," a medical scheduling and doctor booking application graduation project, successfully passing the training program with an 80% evaluation score.

Digital Egypt Pioneers Initiative (DEPI)

Mobile Development Trainee (Flutter)

- Completed training covering Dart, Flutter, Firebase, Git/GitHub, state management, responsive UI/UX, and testing.
- Built mobile applications and a capstone project using Flutter and Firebase.

Information Technology Institute (ITI) - Zagazig

Flutter Mobile Development Trainee

- Studied Dart programming fundamentals, the Flutter framework, responsive UI design, animations, and state management.
- Practiced building application components and integrating with APIs for real-time data handling.

NVIDIA & ITI Mansoura Mentorship Program

Generative AI Trainee

- Selected for a specialized Generative AI program covering Large Language Models (LLMs), prompt engineering, NVIDIA AI tools, and AI fundamentals.

IEEE Zagazig Student Branch (ZSB)

Mobile Development Track Member

- Learned foundational Flutter and Dart, REST API integration, responsive design, animations, and state management.
- Collaborated within a cross-functional team comprising multiple technical tracks to build and deliver a comprehensive final graduation project at the end of the season.

IEEE Brand Ambassadors Program

Mobile Development Mentor

- Mentored students in Flutter and Dart through workshops, technical guidance, and code reviews.

ICPC Zagazig Community

Competitive Programming Trainee

- Completed Level 0 and Level 1 competitive programming training focused on algorithms, data structures, and problem solving.
- Ranked 7th in Level 0 and 4th in Level 1 competitions; solved 533+ Codeforces problems with a maximum streak of 20 days.

Additional Information

Hackathons: NASA Space Apps Challenge Cairo (Developed a weather application using NASA APIs)
Student Activities: Mobile Track Member at Zigzag Student Activity and Zag Systems
Languages: Arabic (Native), English (Professional Working Proficiency)
Military Status: Postponed